United International University

Department of Computer Science & Engineering

Final Examination, Trimester: Spring 2025

Course Code: CSE 3715 **Course Title:** Data Communication

Duration: 2 hours **Total Marks:** 40

(Any examinee found adopting unfair means would be expelled from the trimester/program as per UIU disciplinary rules.)

Answer all the questions

1. The IT department of a university is choosing between two communication systems for voice calls between campuses. One system sets up a dedicated path for each call from start to end, while the other breaks the voice data into small units and routes them independently through the network.

   (a) **Identify** the fundamental differences between the two approaches in terms of setup time, resource allocation, and quality of communication. **[3] (CO1)**

   (b) **Justify** which communication approach is more suitable for voice communication over the internet. **[2] (CO1)**

2. The figure below shows the process of how error detection is handled at different layers of a communication system:

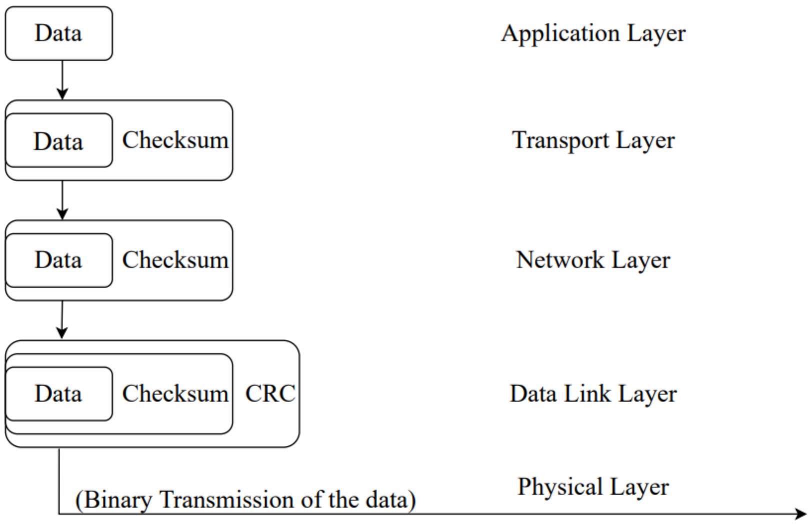

Figure 1: Error Detection Techniques Across OSI Layers

   Assume, the data block size is 3 bits and the data is: `101 011`

   (a) **Identify** the message that the Transport Layer sends to the Network Layer after adding the checksum. **[2] (CO3)**

   (b) **Compute** the final transmitted message after the data from the Transport Layer is encoded using CRC at the Data Link Layer, with generator polynomial, $x^3 + x^2 + 1$. **[5] (CO3)**

(c) Assume the final message is divided into 4-bit blocks and encoded using Hamming code (7,4). At the receiver end, one encoded block is received as 0110111.
Detect and correct any single-bit error in the received block, and **extract** the original 4-bit data. **[5]** **(CO3)**

(d) **Explain** why separate error detection or correction mechanisms are required at both the Transport Layer and the Data Link Layer. **[3]** **(CO3)**

3. In a university building:

- The computer lab uses Ethernet cables to connect all desktops over a wired LAN.
- The library offers wireless access to students using laptops and mobile devices via Wi-Fi.

(a) **Identify** the multiple access control protocol used in wireless networks and **explain** step by step how the protocol operates to manage access to the shared medium. **[5]** **(CO5)**

(b) **Compare** the key differences between wired and wireless multiple access protocols used in these networks. **[3]** **(CO5)**

(c) **Explain** why the multiple access protocol used in wired networks is not suitable for wireless environments. **[2]** **(CO5)**

4. Tanjiro and Zenitsu are having a conversation using a 2G GSM mobile network. Tanjiro is in the Water Hashira's headquarters at Ubuyashiki Village, and Zenitsu is in the Butterfly Mansion in Mt. Sagiri. Both locations operate under the same frequency band allocated by their mobile service provider. During the call, Tanjiro starts traveling by foot toward Mt. Sagiri.

(a) **Describe** briefly how the GSM mobile network connects Tanjiro and Zenitsu and manages their communication. **[4]** **(CO5)**

(b) **Explain** how it is possible for two different cities to use the same frequency band without causing interference. **[3]** **(CO5)**

(c) **Describe** how the network ensures call continuity as Tanjiro moves between different regions. **Explain** various methods for managing the transfer of the call. **[3]** **(CO5)**